

WakeUp! - Real-Time Driver Fatigue & Distraction Monitoring System

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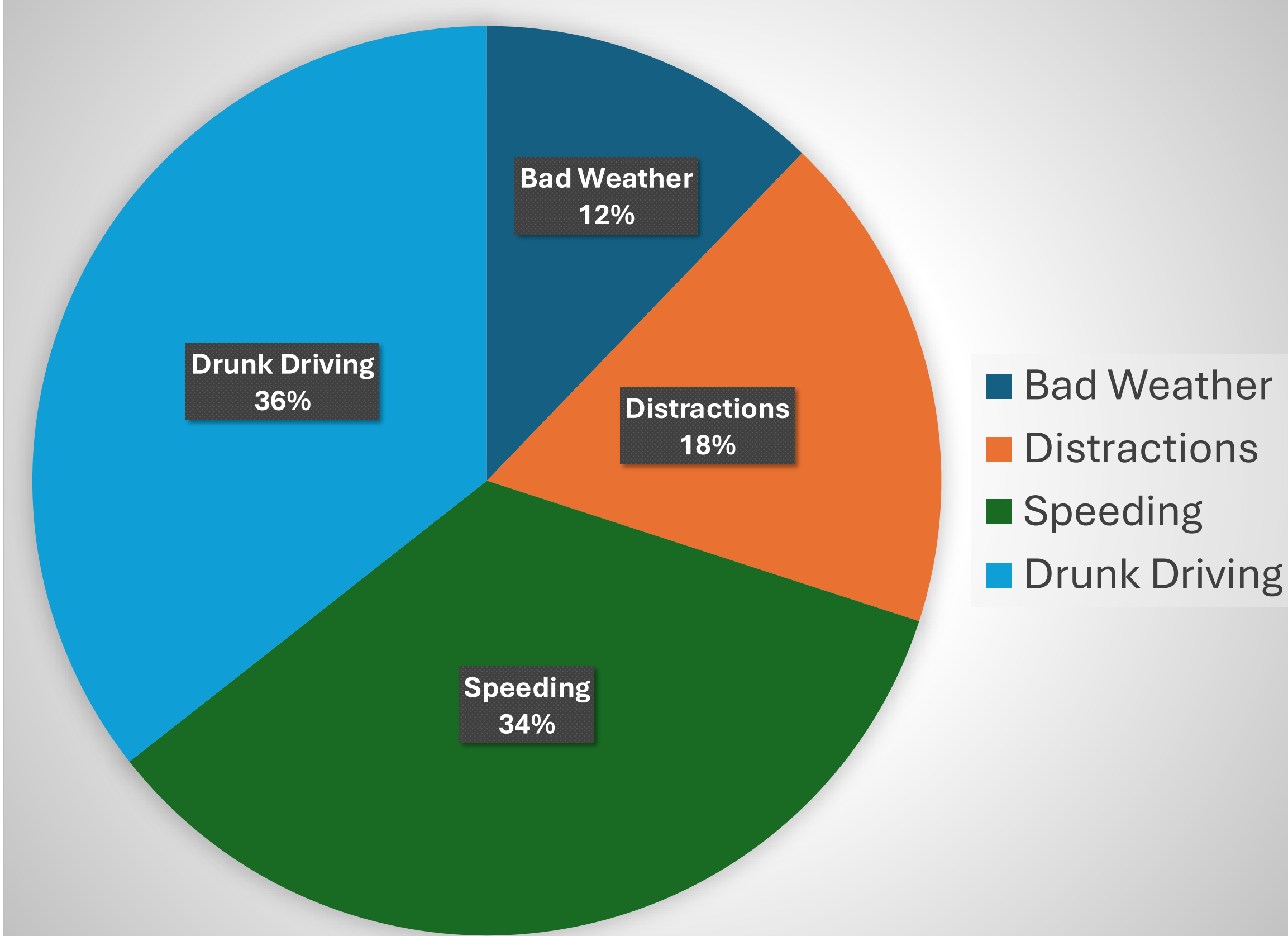
Background & Objective

The Problem: Driver fatigue and distraction cause over 25% of fatal road accidents globally, yet millions of older cars lack integrated safety monitoring.

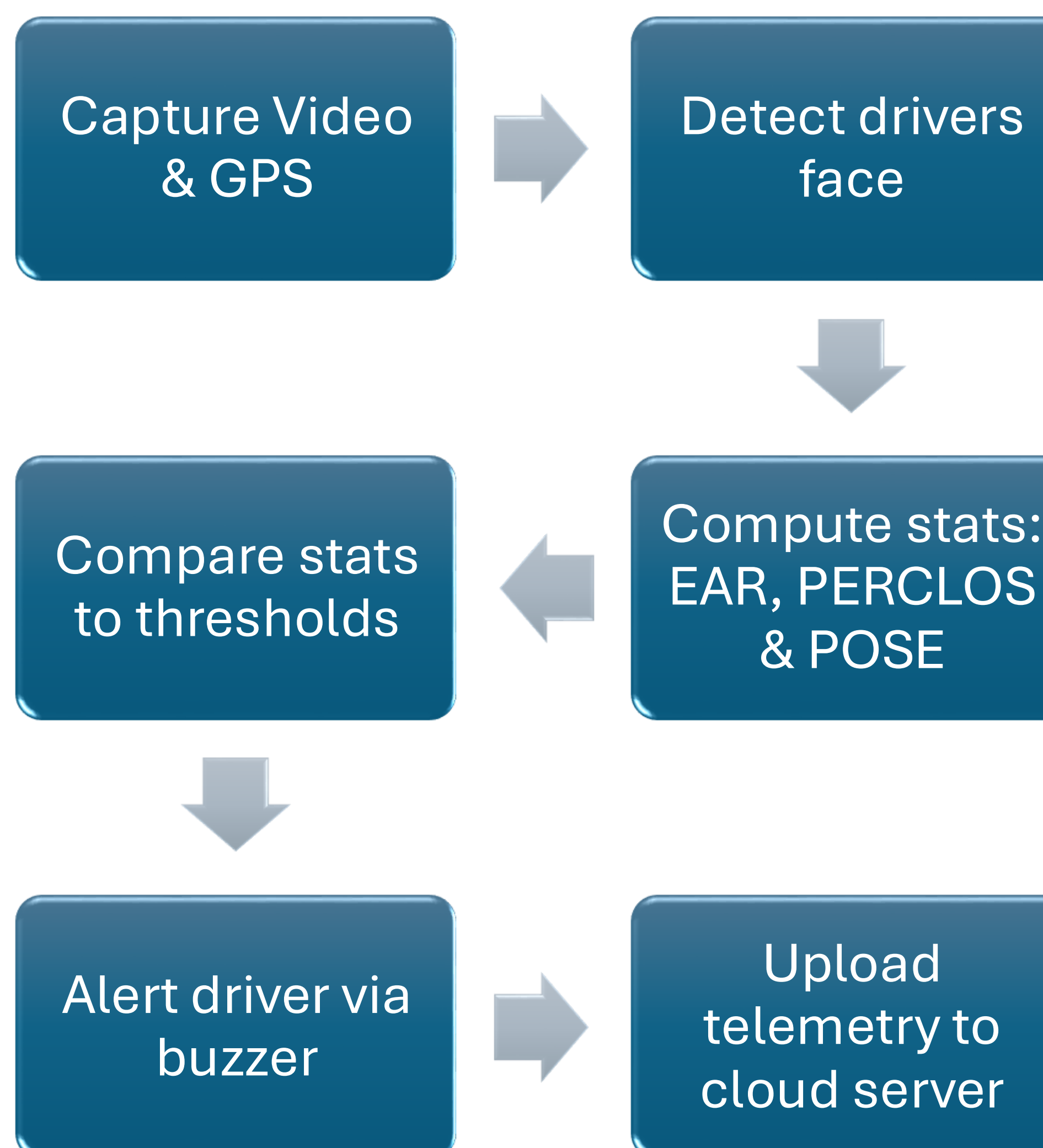
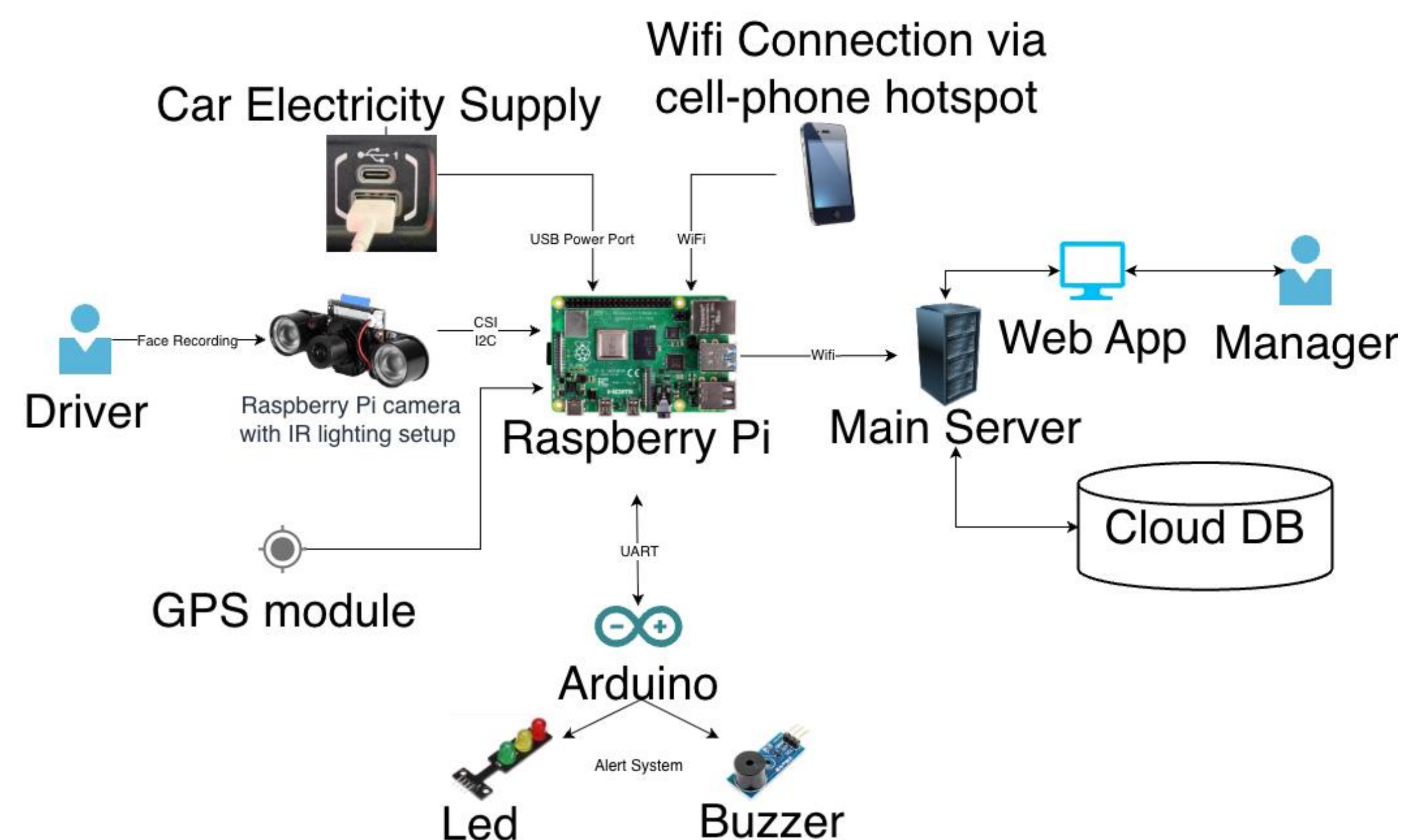
The Solution: WakeUp is a low-cost, retrofittable edge-computing system. It continuously monitors the driver's face using a camera and computer vision.

The Action: Triggers immediate Arduino-controlled buzzer alerts upon detecting dangerous behavior (zero manual interaction) and provides data-driven telemetry to review habits over time.

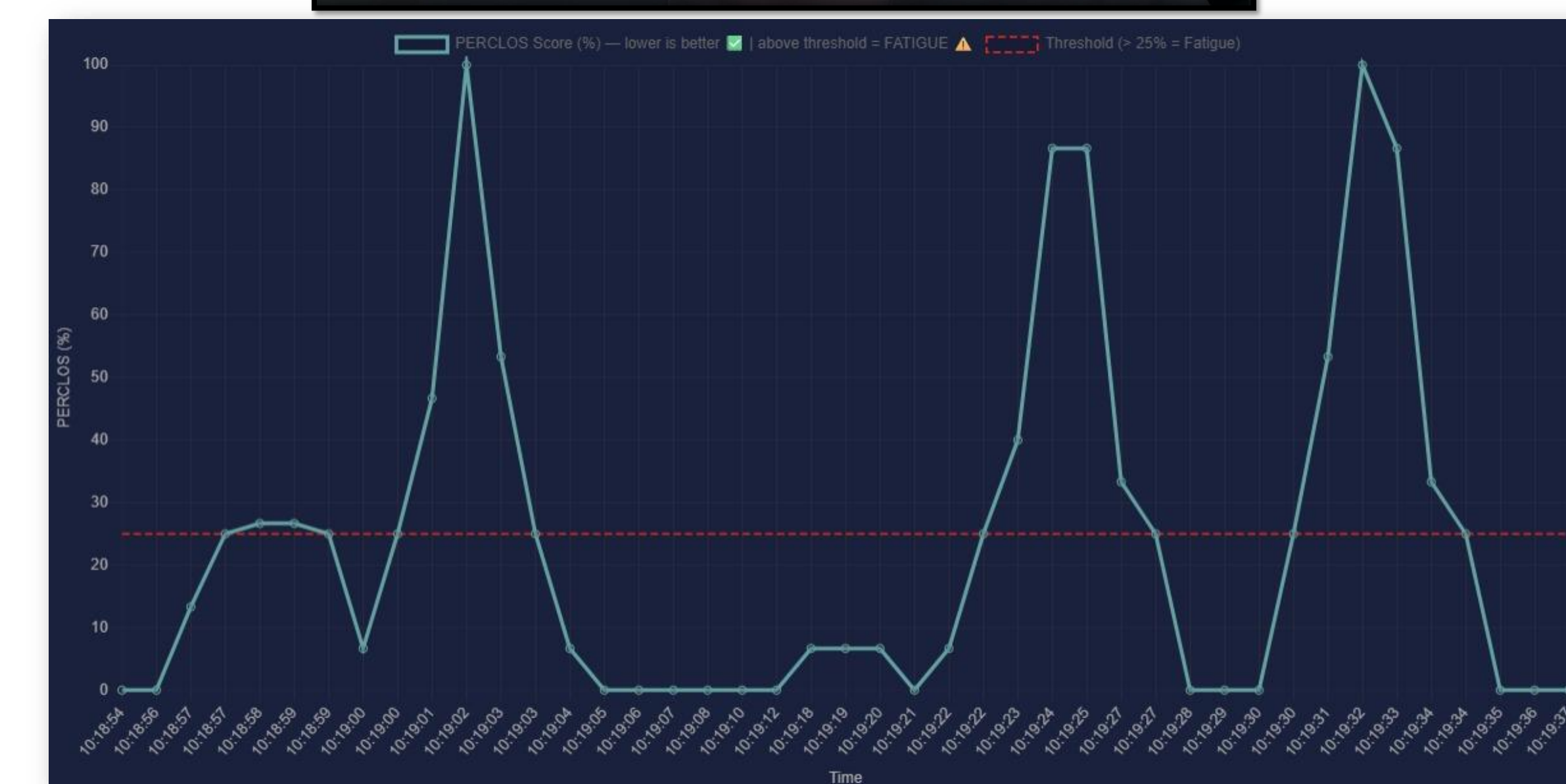
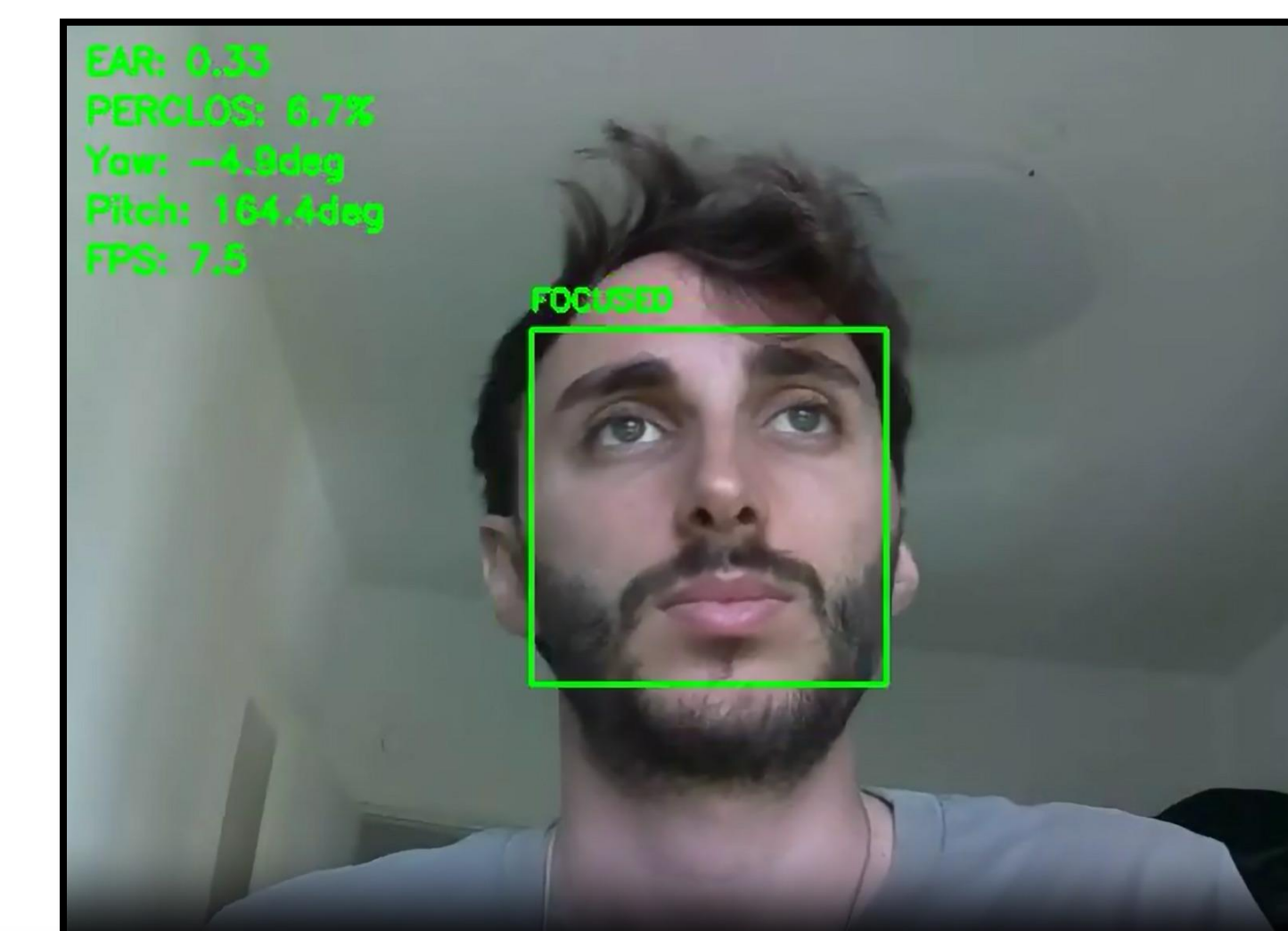
Causes of Car Accidents



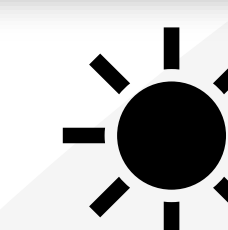
System Architecture



Results



< 1 Sec
Alert latency



85%
Daylight Success Rate



75%
Night Success Rate

Conclusion

Low-Cost Edge Computing: Executes high-accuracy, reliable driver monitoring entirely on a Raspberry Pi 4.

Zero Latency: Requires no internet connection, ensuring instant, life-saving alerts.

Universal Scalability: Offers an affordable, retrofittable safety solution easily installable in any existing vehicle.

